Milk Urea Nitrogen as a Tool to Monitor the Protein Nutrition of Dairy Cows

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ABSTRACT

The relationship between protein nutrition and milk urea N was investigated in three experiments with a total of 125 cows. After 4 wk of pretreatment, cows received 1 of 13 diets with different ratios of protein to energy for 16 wk. Milk was sampled individually for urea analyses during pretreatment and during wk 1, 5, 10, and 15 of treatment. Results were compared with N losses estimated from rumen fermentation and with N losses of metabolic origin.

The mean milk urea N concentration was 12.6 mg/100 ml of milk (range, 9.0 to 18.3 mg). For bulk samples especially, the rumen efflux of crude protein intake was the main determinant of the variation in milk urea N (r^2 = 0.81; residual SD = 1.1). However, N losses from the rumen explained only about 50% of the variation in the milk urea N content of samples from individual cows. The N losses of metabolic origin, which, in these experiments, were responsible for 47 to 100% of urinary N losses, were not related to milk urea N. Results showed that regular measurement of milk urea N in bulk samples can be used to monitor N losses from rumen fermentation. However, the value does not give an indication of the efficiency with which the absorbed protein is utilized.
(**Key words**: cows, milk, urea, protein nutrition)

Abbreviation key: **AP** = true protein digested in the small intestine, **DVE** = AP according to Dutch standard, **DVMFE** = endogenous fecal protein estimated according to Dutch standard, **MUN** = milk urea N, $\mathbf{N_{FPA}}$ = N losses from endogenous fecal protein, $\mathbf{N_L}$ = N losses from milk synthesis, $\mathbf{N_M}$ = N losses from maintenance protein, $\mathbf{N_{REP}}$ = N losses from rumen fermentation, $\mathbf{N_{RET}}$ = N losses from retained protein, **OEB** = REP according to Dutch standard, **REP** = surplus N available in the rumen for microbial growth, **VEM** = Dutch standard for NE_L (1 VEM = 6.9 kJ of NE_L).

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INTRODUCTION

In recent years, the adequate protein and energy nutrition of dairy cows has been increasingly constrained. Higher demands for milk protein relative to other milk constituents puts more emphasis on the adjustment of available protein and energy supply to stimulate efficient milk protein production and synthesis (7, 19). Together with improved genetic potential for milk production, this development has increased the demand for highly digestible roughage and a significant proportion of compound feed in the diet. Highly digestible grass forage can only be obtained after high N fertilization, which not only stimulates forage growth and digestibility but also increases its N content relative to available energy. This increased N may lead to a surplus of N in the diet, lower recovery of N in milk, and higher N losses in urine (7).

This increased demand is, to some extent, contradictory to environmental efforts to reduce avoidable mineral losses and optimize the recycling of nutrients within local farming systems.

The main tools a producer has to avoid N losses are measures to reduce NH_3 emission from the stable, to treat manure, and to balance energy and protein intake as closely as possible to meet requirements. Modern protein evaluation systems [e.g., the one used in The Netherlands (18)] basically provide an opportunity for the producer to monitor protein intake in relation to energy intake and reduce avoidable N losses. Protein evaluation systems distinguish between two main sources of N losses. The first may come from a surplus of N that is available for microbial growth compared with the available energy (**REP**). The excess is absorbed from the rumen and is converted to urea. The second category of N losses originates from true protein digested in the small intestine (**AP**), which is used for maintenance, protein synthesis of milk and tissue, and the replacement of endogenous N losses. In dairy cows, the efficiency of AP utilization for these purposes is about 67, 64, 50, and 67%, respectively, for cows fed diets with the appropriate equilibrium between available protein and energy (18). The remaining protein is converted

TABLE 1. Correlation between blood urea N and milk urea N.

Reference	Correlation
Eckart (4)	0.97–0.99
Piatowski et al. (15)	0.92–0.98
Partschefeld (14)	0.72
Beutlhauser (1)	0.69–0.80
Kaufmann et al. (8)	0.92
Oltner and Wiktorsson (12)	0.98
Oltner et al. (11)	0.91
Lober et al. (9)	0.88

to energy-providing compounds after deamination. The main portion of NH_3 residue is then detoxified in the liver to urea and excreted with urine. In well-balanced diets, the estimated NH_3 losses from AP are quantitatively the most important ones, and the amount becomes higher when intake exceeds requirements (7).

In practice, the appropriate adjustment of protein intake to energy is difficult to realize. Losses during feed storage and dietary selection by the animal might explain why feed analyses are not completely representative of the feed actually consumed. Consequently, other parameters that are easily measurable could be of great practical value as additional indicators for the manipulation of protein supply.

Erbersdobler and Zucker (5) and Oltner and Wiktorsson (12) have postulated that a surplus of N intake increases blood urea N. Moreover, there is a close relationship between blood urea N and milk urea N (**MUN**) (Table 1); therefore, MUN usually increases as the ratio between protein and energy intake increases. Figure 1 presents the pattern of NPN content in bulk samples of milk received from eight dairy operations in The Netherlands over 3 yr. The graphs demonstrate that the NPN content of milk is considerably higher during summer (wk 25 to 45) than during winter, because, in summer, when cows are grazing, the diet mainly consists of grass with a high N content.

A rapid method for the determination of milk urea has become available (3), making possible a regular analysis of bulk milk samples from individual farms or from individual cows. This method may open the possibility of using MUN as an indicator of protein intake in relation to energy intake and the requirements of dairy cows. For that purpose, MUN should not only be an indicator of NH_3 losses from the rumen but also should represent the efficiency of processes that are related to protein metabolism. Insight into metabolic losses may be especially important to the producer. Hof et al. (7) showed that the ratio between AP and NE_L not only influenced milk protein production but also influenced the efficiency of milk protein synthesis and related N losses. However, little information is available concerning the extent to which MUN provides an indication of the N losses assumed to originate either from the rumen or from metabolic origin according to the modern protein evaluation systems. The present study was undertaken to investigate which sources of N losses were reflected by MUN.

MATERIALS AND METHODS

Design of the Experiments

In three experiments with 125 dairy cows, 13 diets were fed with different ratios of **VEM** [Dutch standard for NE_L (21)], **DVE** (AP according to Dutch standard), and **OEB** (REP according to Dutch standard) [Tamminga et al. (18)]. Each experiment started approximately 1 wk after calving with a 4-wk pretreatment period during which time all cows were fed according to protein and energy requirements (2). Thereafter, the 16-wk treatment period began. During this period, diets were offered for ad libitum intake to each individual cow. The mean DVE content in the diets differed among treatments and ranged from 62 to 108 g/1000 VEM. The range of mean OEB intake was –4 to 30 g/1000 VEM (Table 2).

The in vitro procedure of Tilley and Terry (20) was used to estimate VEM, and in sacco procedures (13) were used to measure the protein values for the DVE and OEB of the dietary constituents. Energy intake

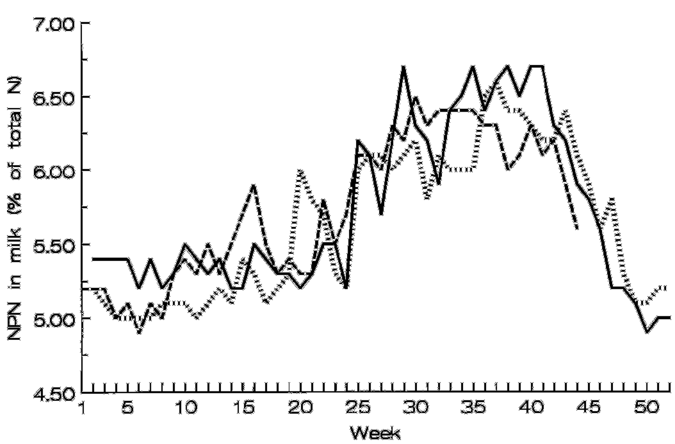

Figure 1. The NPN content in bulk samples of milk delivered to eight dairy companies in The Netherlands (M. C. van der Berg, 1993, personal communication). Legend: 1989 (solid line), 1990 (horizontally dashed line), and 1991 (vertically dashed line).

TABLE 2. Mean intakes of energy and protein of individual treatments.

Treatment	Year	VEM[1]	CP	DVE[2]	OEB[2]
			N		
			(g/1000 VEM)		
1	1991	106.7	142.3	77.5	–1.9
2	1991	108.5	142.8	78.0	–1.8
3	1991	102.7	152.7	86.5	–0.1
4	1991	99.4	154.2	87.7	0.3
5	1991	97.5	162.9	99.9	7.1
6	1992	99.8	144.4	90.9	–3.9
7	1992	98.6	184.1	108.2	19.2
8	1992	110.0	147.6	76.7	6.1
9	1993	110.4	149.0	69.8	13.1
10	1993	124.9	149.5	62.1	19.3
11	1993	96.5	161.2	85.0	13.1
12	1993	102.9	162.1	75.3	21.5
13	1993	100.8	162.9	65.1	29.7

[1]1 VEM = 6.9 × kilojoules of NE_L (21).

[2]DVE = True protein digested in the small intestine; OEB = surplus N available for microbial growth compared with N available for energy, both according to Dutch standards (18).

was expressed as a percentage of the VEM requirement (2) according to individual production. Relative to VEM intake, the DVE intake supplied 75% (treatment 10) to 118% (treatment 7) of the estimated requirement (Table 2). Overall OEB intake was slightly negative in 4 treatments, indicating that, in those treatments, the amount of available N in the rumen was limiting for microbial protein synthesis compared with the amount of available energy. The other diets supplied a surplus of N that was available for microbial growth.

More details concerning diet composition, feed intake, and experimental procedures have been described previously (7).

Milk Sampling for Urea Analysis

In wk 3 and 4 of the pretreatment period and in wk 1, 5, 10, and 15 of the treatment period, morning and evening milk samples were taken from each cow. The samples were frozen until urea analysis. Previous research has shown that this method of conservation did not influence MUN.

Urea was analyzed by the Melkcontrole Station West-Nederland (Gouda, The Netherlands) according to the segmented flow method as described by De Jong et al. (3).

Estimation of N Losses

The losses of absorbed N, either from OEB or from DVE in protein metabolism, were estimated from the individual regression formulas applied in the DVE-OEB system (18). The equations used are given in Table 3.

When OEB was positive, the surplus N ($\mathbf{N_{REP}}$) was assumed to be completely absorbed from the rumen and converted into urea (Table 3; Equation [1]). When the OEB was negative, the DVE intake was corrected for balance.

The DVE required for maintenance, except that lost with hair or scurf ($0.2 \times BW^{0.6}$), was assumed to have contributed to the blood urea N ($\mathbf{N_M}$; Table 3, Equation [2]).

The experiment allowed estimation of the efficiency of milk protein synthesis for each cow during each week (7). These individual efficiency values were then used to estimate N losses related to milk production ($\mathbf{N_L}$; Table 3, Equation [3]).

The DVE-OEB system assumes that 10% of the energy in body reserves consists of energy in protein, which is equivalent to 29 g of protein/1000 VEM. The system further assumes that, during an energy shortage, the mobilized protein fraction is used for milk protein production with an efficiency of 80%. Hence, 20% (5.8 g of protein/1000 VEM mobilized) is degraded to urea ($\mathbf{N_{RET}}$; Table 3, Equation [4]). Energy retention is assumed to be 50% efficient. Thus, when energy balance is positive, 29 g of protein are degraded to urea per 1000 VEM restored (Table 3; Equation [5]).

Finally, the DVE-OEB system corrects for the amount of DVE used to compensate for the endogenous protein losses in the digestive tract (**DVMFE**). Resynthesis of these losses is assumed to occur with 67% efficiency; 33% of this source ($\mathbf{N_{FPA}}$) is converted into urea (Table 3, Equation [6]).

TABLE 3. Equations used to estimate grams of N losses from individual sources.[1]

Source of N	Equation	
Rumen	N_{REP} = OEB/6.25	[1]
Maintenance	$N_M = (DVE_M - 0.2 \times BW^{0.6})/6.25$	[2]
Milk	$N_L = (1 - DVE_{EFF}) \times$ grams of protein/6.25	[3]
Retained	$N_{RET} = -0.0058 \times VEM_{RET}/6.25$ (mobilization)	[4]
	$N_{RET} = 0.029 \times VEM_{RET}/6.25$ (retention)	[5]
Endogenous	N_{FPA} = DVMFE/(0.33 × 6.25)	[6]

[1]OEB = Surplus N available for microbial growth compared with N available for energy, DVE = true protein digested in the small intestine, DVE_M = DVE requirements for maintenance, and DVMFE = endogenous fecal protein according to Dutch standard. All were estimated according to Dutch standards. DVE_{EFF} = Estimated efficiency of milk protein synthesis (7); VEM_{RET} = estimated NE_L retention [kilojoules of NE_L/6.9; (18)].

TABLE 4. Mean production of milk and milk protein and milk urea N content per treatment.

	n	Milk		Milk protein		MUN[1]	
		(kg)		(%)		(mg/100 ml)	
		$\bar{X}$	SE	$\bar{X}$	SE	$\bar{X}$	SE
Treatment							
1	40	31.0^{c}	3.9	3.31^{a}	0.26	9.03	2.05
2	44	30.6^{c}	3.6	3.26^{ab}	0.24	9.13	2.50
3	44	31.9^{bc}	3.8	3.35^{a}	0.20	12.37	2.95
4	44	32.4^{bc}	4.9	3.34^{a}	0.26	11.86	2.47
5	56	35.9^{a}	5.6	3.14^{bc}	0.20	13.67	2.33
6	56	33.8^{ab}	5.0	3.24^{ab}	0.23	9.51	1.68
7	56	36.3^{a}	5.5	3.18^{bc}	0.26	16.55	3.00
8	28	35.5^{a}	4.0	3.08^{c}	0.13	10.45	1.78
9	28	34.2^{ab}	4.4	3.09^{c}	0.20	12.60	1.87
10	28	31.9^{bc}	5.8	3.07^{c}	0.15	13.80	2.10
11	28	34.0^{ab}	4.7	3.15^{bc}	0.19	14.00	2.62
12	24	32.9^{bc}	4.0	3.13^{bc}	0.12	15.45	2.43
13	24	32.2^{bc}	4.8	2.96^{d}	0.12	13.68	2.39
Sampling							
1 wk	125	36.8^{a}	4.4	3.14^{c}	0.20	11.78	3.26
5 wk	125	35.0^{b}	4.3	3.12^{c}	0.20	12.12	3.96
10 wk	125	32.3^{c}	4.3	3.22^{b}	0.23	12.82	3.50
15 wk	125	29.5^{d}	3.9	3.32^{a}	0.26	13.59	3.47

^{a,b,c,d,e,f,g}Means within a column and category with different superscripts differ ($P \le 0.05$).

[1]Milk urea N.

TABLE 5. Mean N losses from separate sources of protein metabolism.

	Estimated losses[1]									
	N_{REP}		N_M		N_L		N_{RET}		N_{FPA}	
	(g of N/1000 VEM[2])									
	$\bar{X}$	SE	$\bar{X}$	SE	$\bar{X}$	SE	$\bar{X}$	SE	$\bar{X}$	SE
Treatment										
1	0.01^{g}	0.03	0.81^{abc}	0.05	3.32^{f}	0.54	0.30^{bc}	0.23	1.10^{a}	0.01
2	0.00^{g}	0.02	0.82^{abc}	0.08	3.22^{f}	0.64	0.34^{b}	0.23	1.10^{a}	0.01
3	0.09^{g}	0.13	0.82^{abc}	0.06	4.43^{d}	0.78	0.24^{bcd}	0.21	1.11^{a}	0.01
4	0.15^{g}	0.15	0.85^{ab}	0.07	4.48^{d}	0.85	0.23^{bcd}	0.20	1.10^{a}	0.01
5	1.31^{e}	0.27	0.82^{abc}	0.06	6.84^{b}	0.61	0.18^{cd}	0.17	0.97^{g}	0.01
6	0.00^{g}	0.00	0.83^{abc}	0.05	5.12^{c}	0.62	0.18^{cd}	0.16	0.98^{fg}	0.02
7	3.05^{c}	0.46	0.81^{bc}	0.05	7.59^{a}	0.88	0.15^{d}	0.14	0.98^{f}	0.01
8	1.03^{f}	0.53	0.79^{c}	0.07	2.58^{h}	0.69	0.31^{bc}	0.22	1.03^{cd}	0.02
9	2.10^{d}	0.48	0.79^{c}	0.06	1.84^{g}	0.56	0.29^{bcd}	0.22	1.04^{c}	0.01
10	3.12^{c}	0.63	0.79^{c}	0.06	0.84^{i}	0.54	0.51^{a}	0.35	1.06^{b}	0.02
11	2.08^{d}	0.61	0.86^{a}	0.08	3.99^{e}	1.05	0.19^{cd}	0.13	1.02^{e}	0.03
12	3.44^{b}	0.56	0.84^{ab}	0.05	2.58^{g}	0.56	0.17^{cd}	0.17	1.04^{cd}	0.02
13	4.76^{a}	0.52	0.83^{ab}	0.05	1.69^{h}	0.53	0.15^{d}	0.14	1.06^{b}	0.01
Sampling week										
1	1.20^{c}	1.38	0.83^{a}	0.08	3.98^{b}	2.20	0.21^{b}	0.17	1.05^{a}	0.06
5	1.30^{b}	1.43	0.80^{c}	0.06	4.25^{a}	2.06	0.20^{b}	0.22	1.04^{b}	0.06
10	1.45^{a}	1.55	0.81^{b}	0.05	4.40^{a}	2.05	0.22^{b}	0.21	1.04^{b}	0.06
15	1.52^{a}	1.60	0.83^{a}	0.05	4.43^{a}	2.02	0.35^{a}	0.24	1.04^{b}	0.06

^{a,b,c,d,e,f,g,h,i}Means within a column and category with different superscripts differ ($P \le 0.05$).

[1]N_{REP} = N losses from rumen fermentation, N_M = N losses from maintenance protein, N_L = N losses from milk synthesis, N_{RET} = N losses from retained protein, and N_{FPA} = N losses from endogenous fecal protein according to Tamminga et al. (18).

[2]Dutch standard for NE_L (1 VEM = 6.9 kJ of NE_L) (21).

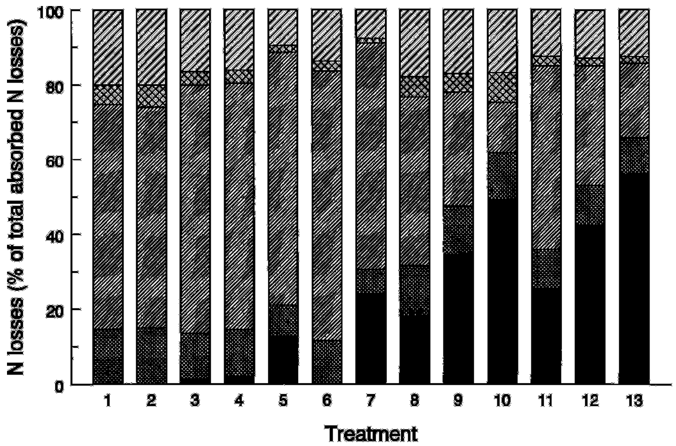

Figure 2. Relative contribution of individual sources to estimated digestible N losses from rumen efflux protein (solid), from absorbed maintenance protein (small crosshatched pattern), from absorbed lactation protein (narrow diagonal pattern), from retained protein (large crosshatched pattern), and from absorbed fecal protein (wide diagonal pattern).

Statistical Analyses

The MUN was more closely related to the ratio between estimated N losses and VEM intake than to absolute N losses. Therefore, comparisons were made between MUN and the estimated grams of N loss per 1000 VEM intake as a variable.

The measurement of MUN as a tool for protein nutrition can be taken in two ways. First, bulk milk from each dairy farm may be analyzed to yield information about the overall ratio of protein intake to energy intake of individual producers. For our data, the bulk samples were simulated by assuming that the mean of each dietary treatment was representative of bulk milk. Second, MUN may be a valuable parameter to monitor protein nutrition of individual cows. This hypothesis was evaluated by analysis of all individual data.

The effect of differences in the ratio of dietary protein to energy on MUN was analyzed using PROC GLM of SAS (16). This procedure was also used for correlation and regression analyses.

No consistent differences in MUN were found between morning and evening samples. Therefore, the means of both were used to estimate the mean content per cow and per week of sampling.

Analyses of the data also showed no systematic relationship between individual cows in MUN content during pretreatment and treatment weeks. From this analysis, it was concluded that there were no genetic differences between cows with regard to MUN. Therefore, only the data obtained from the treatment periods were used for the results.

RESULTS

Means per Treatment

The number of observations, mean milk production, and the protein and urea content in each treatment or each week of sampling are presented in Table 4. Milk production and composition differed ($P \leq 0.05$) between treatments and week of the experiment. Only milk production was affected ($P \leq 0.05$) by age of the cow. The MUN varied from 9.05 to 18.34 mg/100 ml of milk in the bulk samples and from 5.37 to 23.47 mg/100 ml in samples from individual cows. The MUN differed ($P \leq 0.05$) between treatments and weeks. Similar effects of week and treatment were found with estimated mean N losses (Table 5), except for the N_{REP} and N_L, which were not different across weeks.

The relative contribution of individual sources to the overall estimated urea loss varied considerably among treatments (Figure 2). For most treatments, the N loss originated mainly from N_L ($\overline{X}$ = 49.3%; range, 13.4 to 72.1% of total N losses). The N_{REP} represented a smaller proportion ($\overline{X}$ = 20.5%; range, 0 to 56%). Other sources contributed less to the formation of urea; 14.9, 11.6, and 3.7% from N_{FPA}, N_M, and N_{RET}, respectively.

The mean MUN concentration was significantly correlated with the mean values of total N loss (r = 0.96), N_{REP} (r = 0.86), and N_{RET} (r = –0.51). The OEB intake (N_{REP}) was responsible for 81% of the variation in MUN (Figure 3). The N_{REP} plus N_{RET} explained 88% of the variation, and total N loss explained 86% of the variation (Table 6).

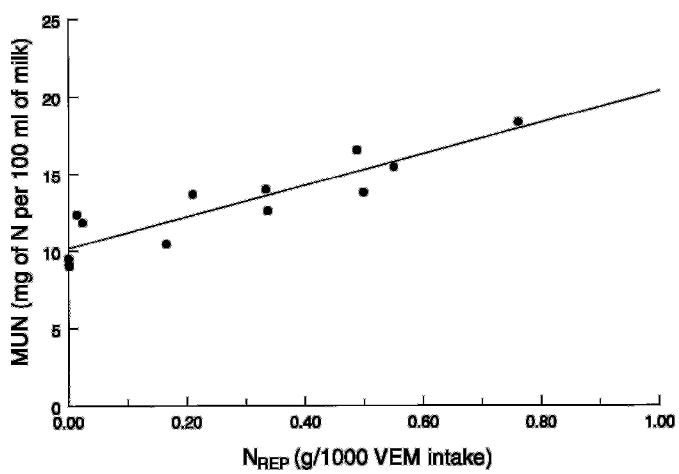

Figure 3. Influence of the intake of the rumen efflux protein on mean milk urea N (MUN) content. N_{REP} = N losses from rumen fermentation; VEM = Dutch standard for NE_L (1 VEM = 6.9 kJ of NE_L).

These results strongly suggest that MUN of bulk samples per operation can serve as an indicator of the rumen efflux protein. If OEB = 0 (N and energy available for microbial synthesis in balance), mean MUN was estimated to amount to 10.3 ± 0.5 mg/100 ml of milk. The concentration would increase by 1.6 ± 0.2 mg/100 ml if OEB intake increased 1 g/1000 VEM intake.

Individual Cows

The MUN concentrations of samples from individual cows were much less indicative of the protein status of an individual cow than were the mean MUN concentrations per operation. Total N loss from DVE (grams per 1000 VEM) explained 52% of the variation in MUN (Table 6). Although four potential sources of these losses were significantly correlated with MUN, those with N_{REP} were mainly determinant. The inclusion of other factors in the equation (e.g., N losses from other sources, lactation, week, or age) made little or no contribution to the estimate.

Also, based on individual data, MUN should have amounted to 10.3 ± 0.1 mg/100 ml of milk when OEB = 0. The concentration should increase by 1.7 ± 0.1 mg/100 ml per g of OEB/1000 VEM intake.

DISCUSSION

In The Netherlands, the mean annual N input on dairy farms is 175 kg per cow. Almost 20% of this N is recovered in milk, 30% is excreted in feces, 2% is retained in tissues, and 50% is lost with urine (17, 22). A considerable part of the excreted N is lost by NH_3 volatilization and nitrate leaching, both of which contribute to environmental deterioration through eutrophication and acidification. The amount of NH_3 from cattle husbandry in 1990 (140 thousand tonnes) was estimated to amount to 55% of total NH_3 emission in The Netherlands (10). According to Dutch legislation, NH_3 emission in 2000 should be reduced by 50% compared with NH_3 emission in 1980. Therefore, a practical tool that would inform the producer regularly about the degree to which protein intake is tuned to protein requirement would be of great value.

The results of this experiment confirm previous work (4, 5, 6) that showed that a significant relationship between MUN and the efficiency of protein utilization by the cow. Oltner and Wiktorsson (12) postulated that MUN was more closely related to the ratio of dietary protein to energy than to absolute protein intake. Our results agreed with that conclusion concerning the ratio between these nutrients available for microbial protein synthesis.

More than 80% of the variation in MUN in bulk samples was related to N losses from rumen fermentation (OEB) and metabolism (DVE). However, only the relationship between MUN and OEB intake per 1000 VEM was significant. This relationship was less pronounced for individual cows for which the estimated N losses explained only 50% of the variation in MUN. For individual cows, the ratio of OEB to VEM was also the main factor responsible for the variation in MUN.

TABLE 6. Relationships between milk urea N and protein nutrition.

	N Losses								
	N_{REP}	N_M	N_L	N_{RET}	N_{FPA}	N_T	C	R^2	RSD
	(g/1000 VEM[2])								
Treatment correlation	0.90*	0.23	–0.07	–0.40	–0.28	0.92*			
Regression[3]									
1						1.8*	1.7	0.86	1.1
2	1.6*						10.2*	0.81	1.3
3	1.5*			–7.8*			12.4*	0.88	1.1
Correlation per individual cow	0.70*	–0.02	0.12*	–0.06	–0.29*	0.72*			
Regression[3]									
1						1.6*	2.7*	0.52	2.5
2	1.7*						10.3*	0.49	2.6
3	1.6*				–3.8		14.3*	0.49	2.6
4	2.0*		17.4*		12.7*		–6.8*	0.57	2.4

[1]N_{REP} = N losses from rumen fermentation, N_M = N losses from maintenance protein, N_L = N losses from milk synthesis, N_{RET} = N losses from retained protein, N_{FPA} = N losses from endogenous fecal protein, N_T = N losses from total absorbed protein.

[2]Dutch standard for NE_L (1 VEM = 6.9 kJ of NE_L) (21).

[3]Milk urea N = $a \times N_{REP} + b \times N_M + c \times N_L + d \times N_{RET} + e \times N_{FPA} + f \times N_T + C$. Milk urea N was measured in milligrams of N/100 ml of milk, and variables were measured in grams of N/1000 VEM.

*$P \le 0.05$.

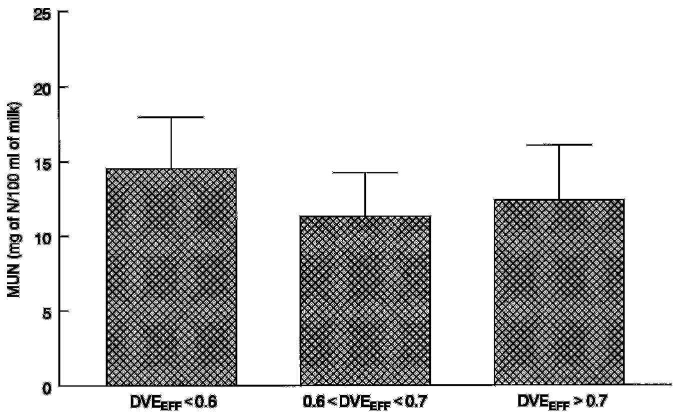

Figure 4. Milk urea N (MUN) in relation to the efficiency of utilization of true protein digested in the small intestine (Dutch standard) for milk protein synthesis. DVE_{EFF} = Estimated efficiency of milk protein synthesis (7).

The MUN was not clearly affected by the ratio between N_L and VEM intake. This result is remarkable because the mean metabolic N losses from lactation protein were estimated to represent at least 80% of the total losses of absorbed N in these experiments and were not confounded with those from OEB intake. Part of the N losses from protein metabolism is recycled into the rumen and is used for microbial growth. The extent to which that occurs is mainly determined by OEB intake. Higher OEB intake leads to less use of the recycled endogenous urea and more excretion with urine and milk. Because of these dynamics of urea in the lactating cow, urea concentrations in milk do not clearly reflect the possible dietary limitation of absorbed protein.

A correct balance between DVE and energy intake is essential for milk protein production and the efficiency of milk protein synthesis. A surplus of DVE in relation to energy that is available for milk protein production leads to inefficient synthesis, which is quantitatively one of the most important sources of dietary N losses (7). Therefore, monitoring metabolic losses from DVE is at least as important as monitoring OEB intake. Figure 4 demonstrates the mean MUN concentration from cows with relatively low (<60%), average (≤60% but ≤70%), or high (>70%) DVE efficiency for milk production, compared with the mean efficiency of 64%, which is the standard in the DVE-OEB system. Figure 4 clearly indicates that, in these experiments, there was no relationship between the calculated efficiency of milk protein synthesis and MUN. Only a small, nonsignificant trend could be seen whereby MUN increased when the DVE efficiency was <60 or >70%.

Regular analyses of the MUN as a tool to monitor N losses is only valuable if the concentration yields reliable information about the causes for the losses, but the results of this study clearly showed that this information was not always reliable. The MUN was only valuable as an indicator of OEB intake. Moreover, reliable information can only be obtained from bulk samples.

From these results, it can be concluded that MUN concentration was representative of the surplus of N for microbial synthesis in the rumen (OEB). The MUN amounted to ca. 10.3 mg/100 ml when diets with an OEB value = 0 were fed. This content increased by 1.7 g per additional gram of OEB per 1000 VEM intake. However, the parameter cannot be used as an indicator for the utilization of absorbed true protein and, therefore, cannot be used to monitor the ratio between DVE and VEM intake. Analysis of bulk samples from a dairy operation yields more reliable information on N losses because of OEB intake than do samples from individual cows.

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